

Clearing the Air: How Pollution Shapes Labor Supply in Bangladesh

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Research on the impact of air pollution on labor supply, particularly in rural areas of developing countries, is considerably limited. This paper examines the impact of air pollution on hours worked in rural Bangladesh. Utilizing panel household survey and satellite data on pollution, this study employs an instrumental variable approach that exploits the exogenous occurrence of thermal inversions. A 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ reduces monthly hours worked by 18.8% (20.7 hours per month) relative to the mean of 109.8 hours. The impact is concentrated among older workers although it does not differ significantly by gender or household income. Overall, these findings underscore the labor-market costs incurred in highly polluted, low-income settings where such effects remain understudied.

Keywords: pollution, hours worked, labor supply

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1 Introduction

In recent years, air pollution has emerged as a pressing global environmental issue and a major threat to human health. Globally, nearly 99% of the population breathes air containing pollutants exceeding World Health Organization (WHO) recommended thresholds, contributing to nearly 6.7 million deaths annually (WHO 2024). In 2019 alone, pollution caused 4.2 million premature deaths, of which 89% occurred in low- and middle-income countries (LMICs) (WHO 2024). The situation is particularly worse in LMICs where rapid development and urbanization have rendered air unbreathable (Brooks et al. 2023; Jayachandran 2022; Dechezlepretre, Rivers and Stadler 2019).

Air pollution worsens health (Currie et al. 2014; Tanaka 2015), educational performance (Balakrishnan and Tsaneva 2021; Bedi et al. 2021; Bharadwaj et al. 2017; Ebenstein et al. 2016; Lavy, Ebenstein, and Roth 2014; Roth 2021; Zhang et al. 2018; Zivin et al. 2020), subjective well-being (Li et al. 2019, Zheng et al. 2019) and labor market productivity (Adhvaryu, Kala, and Nyshadham 2022; Chang et al. 2019; Chen and Zhang 2021; Li, Liu, and Salvo 2015; Zivin and Neidell 2012; Fu et al. 2017; He et al. 2019). Despite such growing evidence, much less is known about how pollution affects labor supply in rural areas where employment is often informal and physically intensive, thereby rendering labor markets more vulnerable to such environmental shocks. This paper therefore investigates the impact of pollution on rural labor supply in Bangladesh, the most polluted country in the world. This research contributes to two strands of economic fields: labor and environment.

The Bangladeshi context is relevant to examine the relationship between air pollution and labor supply. Even though the country has witnessed sustained economic growth over the past two decades, it remains the world’s most polluted country (Energy Policy Institute 2023) with 164.8 million people living in areas where average pollution levels exceed both the WHO guideline of $5 \mu\text{g}/\text{m}^3$ and the national standard of $35 \mu\text{g}/\text{m}^3$. Air pollution is the second greatest threat to human health in the country, with the economic cost of health issues attributable to pollution ranging between US\$11.5–13 billion, or 3.9-4.4% of GDP in 2019

compared to past year (Raza, Mahmud, and Rabie 2022). These costs reflect productivity losses due to impaired health, reduced labor supply and the existence of pollution-based poverty traps, where high pollution levels exacerbate health, education and labor market outcomes (Fuller et al. 2022).

In Bangladesh, rural areas are more vulnerable to the adverse impact of pollution on labor markets. Households located in these areas are ill-equipped to buffer themselves against environmental shocks. The lack of adequate health infrastructure coupled with insufficient information on the health risks of pollution reduces the likelihood that individuals get treated on time (Balakrishnan and Tsaneva 2021). The problem is further compounded with workers in rural areas more likely to engage in physically demanding jobs such as manufacturing and agriculture, due to low education and limited technical skills (Huang et al. 2021). Moreover, the lower baseline health status common in developing and low-income countries exacerbates pollution impacts (Balakrishnan and Tsaneva 2021; Chen et al. 2013). Prior literature on the labor supply effects of pollution is scarce for developing countries, let alone concentrated only in rural areas (Chan, Pelli, and Vienne 2023; Huang et al. 2021). In this setting, this paper fills an important gap, hypothesizing that pollution impacts are likely to be more severe in rural areas (Huang et al. 2021; Balakrishnan, Cohen and Smith 2014).

Availability and access to reliable pollution data is also a challenge. Nearly 25% of countries having the highest levels of pollution have sparsely located pollution monitors, mostly in urban areas, leaving rural areas unmeasured (Barwick et al. 2019; Fan and Grainger 2023; Balakrishnan et al. 2014; Balakrishnan and Tsaneva 2021). In Bangladesh, the Department of Environment uses Continuous Air Monitoring Station (CAMS) to collect real-time measurement of pollutants in only 8 major cities (Dhaka, Narayaganj, Ghazipur, Chittagong, Rajshahi, Khulna, Barisal and Sylhet) of the country. This problem is further compounded in instances where local officials strategically avoid pollution hotspots when placing monitors (Grainger and Schreiber 2019), increasing pollution emissions during non-monitoring periods (Zou 2021) or manipulate data to report lower pollution levels (Gu, Bian, and Elahi 2020).

To counter these problems, I utilize satellite data instead (Jayachandhran 2009). I utilize pollution data from Atmospheric Composition Analysis Group (ACAG) at Washington University in St. Louis (van Donkelaar et al. 2021), which provides monthly pollution levels at a spatial resolution of 1 km. Using household coordinates, I match each household to the pollution grid cell and merge them with BIHS survey data.

The identification in this paper relies on variation in pollution over time and across households in Bangladesh. As pollution is not randomly assigned across space or time, it is difficult to establish a causal link between pollution and labor supply. Potential sources of endogeneity include residential sorting, reverse causality and environmental confounding (Freeman et al. 2019; Khanna et al. 2021; Zivin and Neidell 2013; Aguilar-Gomez et al. 2022). Conditional on weather controls, thermal inversions, when a mass of cold air gets caught below a mass of hot air, trapping pollutants and degrading air quality, provide a plausibly exogenous source of variation in pollution, making them a suitable instrumental variable to address these concerns. The identifying assumption of my instrument is that, conditional on district fixed effects, season by year fixed effects, weather and household controls, changes in thermal inversions are unrelated to changes in labor outcomes except through their impact on air pollution. My first stage estimates show that there is a positive and statistically significant relationship between thermal inversions and air pollution, with a one standard deviation increase in thermal inversions increasing PM2.5 concentrations by $2 \mu\text{g}/\text{m}^3$ (2.2%). My baseline results indicate that a $10 \mu\text{g}/\text{m}^3$ increase in PM2.5 leads to a reduction of about 20.7 hours (18.8%) worked by rural workers per month. My results are consistent across alternative specifications. In particular, my robustness checks include controlling flexibly for weather, using alternative instruments, restricting the working sample, conducting a placebo test and winsorizing both pollution and hours worked to mitigate the influence of extreme values. Across all specifications, my estimates remain robust and consistent.

This paper contributes to the existing literature on air pollution and labor markets in

three main ways. First, it contributes to the quasi-experimental literature on the impact of air pollution on labor supply. Most of the existing studies focus on determining countrywide or urban labor supply responses or productivity outcomes, with little focus on rural areas in developing countries (Carson, Koundouri, and Nauges 2011). I also utilize high resolution satellite data on pollution and thermal inversions which allows me to overcome sparse coverage, urban bias and potential manipulation concerns associated with developing country ground-based monitoring data. Second, I utilize household panel data with a comprehensive set of weather and demographic controls and a rich fixed effects structure to address unobserved time invariant and time varying factors that could affect both air pollution and labor supply. Leveraging thermal inversions as an exogenous source of variation in pollution allows me to address key identification concerns discussed earlier and yield credible causal estimates.

The remainder of the paper is organized as follows: Section 2 provides a brief background on air pollution and its impact on health and labor supply. Section 3 describes the data sources and presents the descriptive statistics. Section 4 provides the empirical framework, explains the identification strategy and presents the main instrumental variable results. Section 5 concludes the paper.

2 Context

2.1 Air Pollution in Bangladesh

Considered as the most polluted country in the world, Bangladesh has 164.8 million people residing in areas where the particulate pollution exceeds the country’s national standard of $35 \mu\text{g}/\text{m}^3$ (Energy Policy Institute 2023). The country’s residents are expected to gain 6.7 years in life expectancy if this guideline is met. In 2019, air pollution was considered to have led to 78000-88000 deaths and 1 billion days lived with illnesses in the country (Raza, Mahmud, and Rabie 2022). In terms of economic losses, air pollution led to a fall in GDP

by 3.9-4.4% in 2019 (Raza, Mahmud, and Rabie 2022). The major sources of air pollution include vehicular emissions, coal-fired power plants, brick kilns, construction activities and biomass burning.

By 2023, the country’s annual average PM_{2.5} concentration was 79.9 $\mu\text{g}/\text{m}^3$, more than twice the national standard set in 2022 (35 $\mu\text{g}/\text{m}^3$) and 15 times the WHO’s guideline of 5 $\mu\text{g}/\text{m}^3$. If the PM_{2.5} concentrations were reduced to meet the national standard, 19% deaths, 21% Years of Life Lost (YLL) and 12% Years Lived with Disability (YLD) could be averted (Nesan and Kelly 2025). In 2023 alone, pollution was responsible for 102,456 deaths from ischemic heart disease, stroke, chronic obstructive pulmonary disease, lower respiratory infections and lung cancer (Nesan and Kelly 2025). High pollution also led to significant morbidity with 670, 000 annual emergency room visits due to asthma and 262 million lost workdays annually.

Among particulate matter, PM_{2.5} is considered be more detrimental to health than other coarser particulate matter (Hoffmann and Rud 2022). Consisting of tiny particles having a mean aerodynamic diameter of up to 2.5 μm , PM_{2.5} can remain airborne from a few days to weeks and penetrate deep into the lungs and impair respiratory function (Aguilar-Gomez et al. 2022; Chay et al. 2003; Isen et al. 2017). It is emitted by natural sources such as volcanoes and wildfires or anthropogenic sources such as industrial activities, combustion of fossil fuels including biomass burning and chemical reactions of air emissions (Yamada and Narita 2025).

2.2 Air Pollution and Health Outcomes

Exposure to air pollution can lead to both mild and life-threatening health issues. Mild impacts include changes in blood pressure, headaches, and irritation of the eyes, ear, nose, throat, skin and lungs (Auchincloss et al. 2008; Pope 2000; Ghio, Kim and Devlin 2000; Zhou and Zhang 2023). Pollution is also associated with reduced blood flow and cell oxygenation, leading to chest pain and tightness, fatigue, irritability, and impatience (Huang et al. 2021).

This further impairs attention, productivity and cognitive function, thereby affecting a range of behavioral and socioeconomic outcomes (Aguilar-Gomez et al. 2022).

Furthermore, air pollution has also been linked to chronic illnesses such as diabetes, autism, neurological disorders, and weakened neonatal immune system (Pearson et al. 2010; Morales-Suarez-Varela, Peraita-Costa, and Llopis-Gonzalez 2017; Han et al. 2023). It has also been associated with changing emotional states and causing anxiety and depression (Ehsanifar et al. 2019; Fonken et al. 2011; Davis et al. 2013). Pollution can also lead to cardiovascular and respiratory impairment including chronic coughs, shortness of breath and asthma attacks (Zhang, Hao and Lu 2018; Han et al. 2023). Particulate matter can also penetrate the bloodstream and cause cardiovascular complications such as stroke and heart failure, leading to increased hospitalizations and mortality rates (Bell et al. 2004; Brook et al. 2010; Hoffmann and Rud 2022; Ranft et al. 2009; Calderon-Garciduenas et al. 2014; Genc et al. 2012).

2.3 Air Pollution and Labor Supply

There is a sizable body of evidence focusing on determining the impact of pollution on labor supply responses. Exposure to pollutants has a detrimental impact on workers' labor market outcomes in two main ways. First, it reduces workers' functionality to work, also known as 'presenteeism' and health problems arising from pollution can interfere with workers' daily routines (Chang et al. 2016; Han et al. 2023; Kamble 2024). Second, it can lead to absenteeism with workers missing work or reducing hours worked due to their own illness or to provide caregiving to sick household members (Chang et al. 2016; Hanna and Oliva 2015; Han et al. 2023; Yamada and Narita 2025; Kögel 2023). Studies have found that households having susceptible members including children and the elderly face more detrimental consequences of pollution (Aragon, Miranda, and Oliva 2017; Li and Li 2022). Intra-household heterogeneity reveals mothers reducing their hours worked while fathers' labor supply remains unchanged as the former take on traditional caregiving roles (Han et

al. 2023).

Prior literature focuses both on the extensive margin of labor supply, with pollution leading to reduced labor supply due to its impact on health (Kim, Manley and Radoias 2017) and on the intensive margin, with pollution leading to reduced productivity (Zivin and Neidell 2012; Chang et al. 2016; Archsmith, Heyes and Saberian 2018; Chang et al. 2019; He, Liu and Salvo 2019; Kahn and Li 2020). This includes papers investigating detrimental impact of PM2.5 on hours worked in developing countries (Aragon, Miranda, and Oliva 2017; Hanna and Oliva 2015; Zhang et al. 2018).

In some instances, the relationship between pollution and labor supply is not linear, with research finding a positive relationship between PM2.5 exposure and hours worked upto a threshold beyond which the trend reverses (Zhang, Chen, and Zhang 2018). Furthermore, there is limited evidence on the impact of pollution on labor supply in rural areas (Chan, Pelli, and Vienne 2023). Research also finds that effective legislation can help reduce pollution levels and improve labor supply (Hanna and Oliva 2015; Viard and Fu 2015).

3 Data and Descriptive Statistics

This section outlines the data utilized in this paper. I gather information from multiple data sources to conduct the analysis. Data on labor supply is obtained from Bangladesh Integrated Household Survey (BIHS) 2011-12, 2015 and 2018-19 while pollution data is sourced from Atmospheric Composition Analysis Group (ACAG) at Washington University in St. Louis (van Donkelaar et al. 2021). Data for thermal inversions is obtained from Modern-Era Retrospective Analysis for Research and Applications (MERRA2) and weather data is acquired from ECMWF ERA5 reanalysis data.

3.1 Bangladesh Integrated Household Survey

Conducted by International Food Policy Research Institute (IFPRI), Bangladesh Integrated Household Survey (BIHS) is the most comprehensive, nationally representative individual level rural panel survey in the country. It covers detailed information on 6500 households in 325 primary sampling units regarding key socio-economic indicators including demographic characteristics, education, employment, anthropometric measurements, agricultural production, time allocation and women’s empowerment. Figure 1 plots the location of households included in the sample.

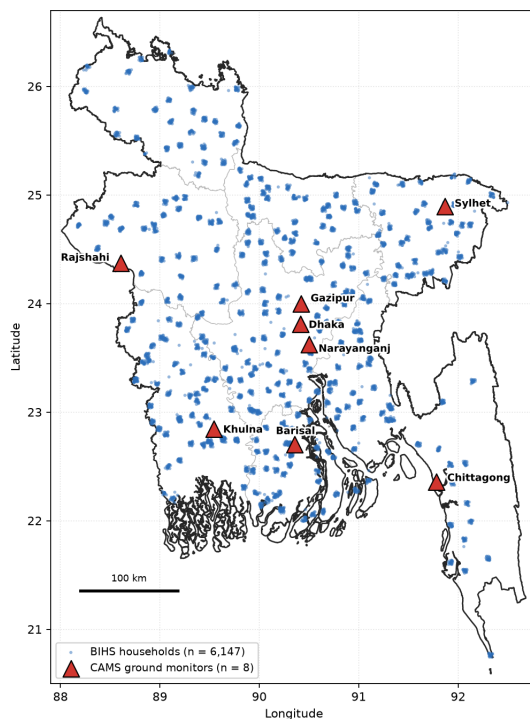


Figure 1: BIHS Household Sample and Ground-Monitor Coverage

Note: The map plots the 6,147 BIHS working-sample households (blue) — individuals aged 15–64 with a primary work activity and complete controls, across the three survey waves (2011–12, 2015, 2018–19) — together with the eight Department of Environment Continuous Air Monitoring Stations (CAMS; red triangles) in Dhaka, Narayanganj, Gazipur, Chittagong, Rajshahi, Khulna, Barisal, and Sylhet. The household sample covers rural areas nationwide, whereas the ground-monitoring network is sparse and concentrated in a few cities — motivating the use of the fine-resolution satellite (ACAG) pollution data. Division boundaries are shown in grey.

The primary variable of interest is the number of hours worked per month. The survey records employment information on all types of economic work performed by each household

member aged 6 years and above in the week preceding the interview. This is the reference period week t . I restrict the sample to working individuals lying in the age range 15-64 years. Each respondent is required to report all economic activities that they are involved in. In cases where more than one activity is reported, I consider the activity with the highest number of hours worked reported as the primary one. I then calculate monthly hours worked. To exclude the influence of outliers, I trim the top 1% of my sample from the monthly hours distribution.

3.2 Pollution Data

I obtain satellite-based pollution data from Atmospheric Composition Analysis Group at Washington University in St. Louis (version V5.GL.05.02) (van Donkelaar et al. 2021). This provides information on monthly mean pollution for each $0.01^\circ \times 0.01^\circ$ grid (1 km at the equator) across the globe. Using household coordinates, I match every household to the pollution grid cell in which it is located and assign it the pollution level for that cell. I calculate average monthly pollution for each household in the sample.

Figure 2 shows the spatial distribution of average monthly pollution levels across the 3 Bangladesh Integrated Household Survey (BIHS) waves (Wave 1: Sept 2011-March 2012, Wave 2: Dec 2014-Aug 2015, Wave 3: Oct 2018-Apr 2019) using Atmospheric Composition Analysis Group data. Pollution values mostly range between 50-120 $\mu\text{g}/\text{m}^3$, far above the national standard set in 2022 of 35 $\mu\text{g}/\text{m}^3$ and more than 20 times the WHO's guideline of 5 $\mu\text{g}/\text{m}^3$. This highlights the severity of pollution and its persistence in the country over the years. During the first wave, pollution is mostly concentrated in the southwestern and central regions of the country. However, it reduces considerably by 2015 although increased PM2.5 concentrations are present in the southwestern regions. In the last wave, pollution resurges, with the highest increases again experienced in the southwestern regions and moderate increases observed in the central regions of the country.

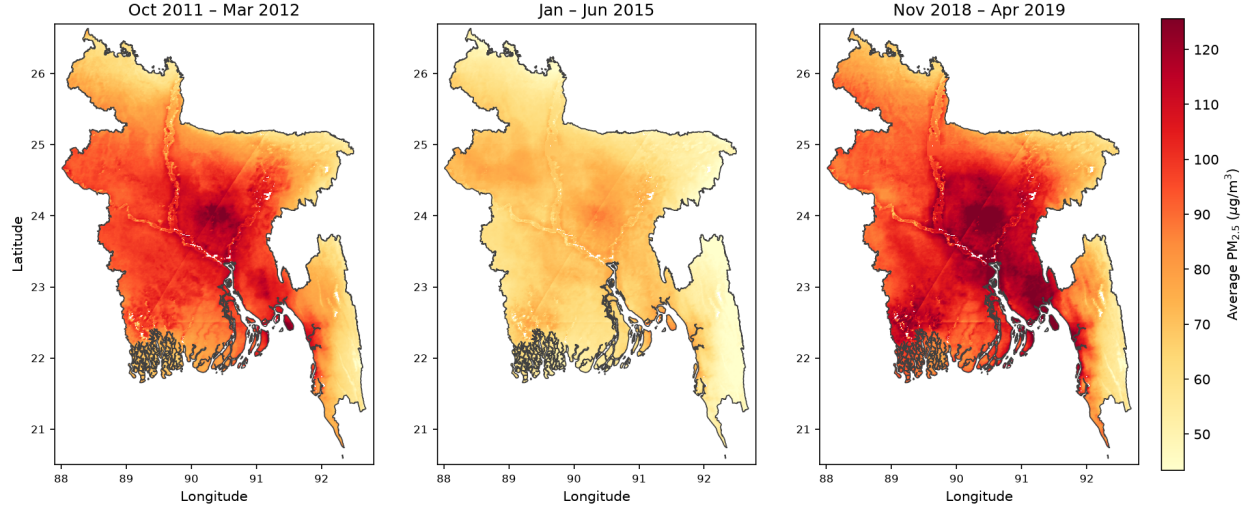


Figure 2: Average PM_{2.5} in Bangladesh (2011–2019)

Note: Average PM_{2.5} over each survey window — Oct 2011–Mar 2012, Jan–Jun 2015, and Nov 2018–Apr 2019 — from the ACAG satellite grids (0.01°; van Donkelaar et al. 2021).

Figure 3 represents the trends in average monthly pollution in Bangladesh over the years 2011–2020. The figure represents the seasonality of PM_{2.5} levels across the year, with strong peaks reached during the dry winter months (October–March) and significant declines during the monsoon season (June–August). Because of this, I include season by year fixed effects in my main regression to account for seasonal variation in pollution.

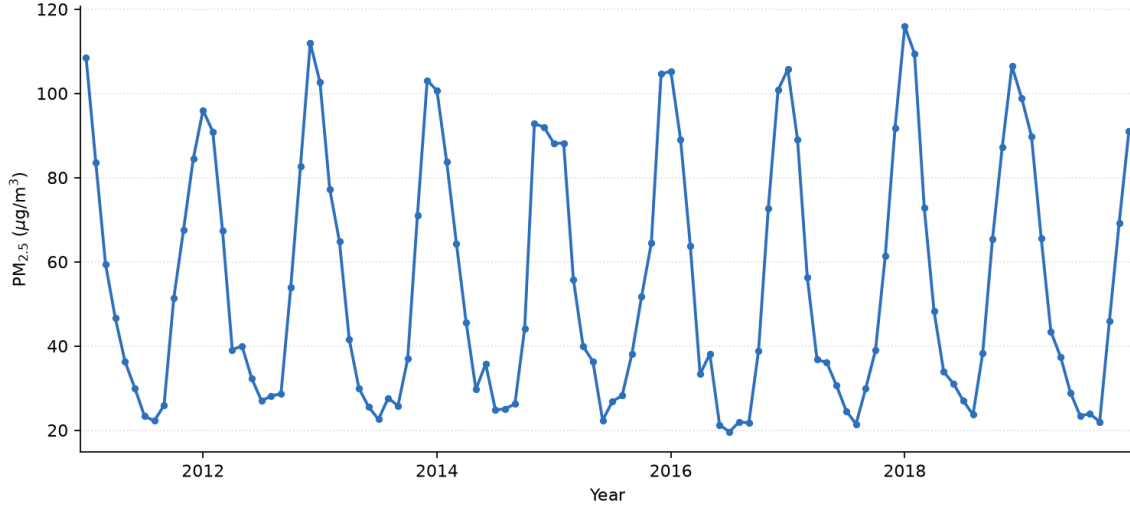


Figure 3: Monthly Average PM_{2.5} in Bangladesh, 2011–2019

Note: Monthly mean PM_{2.5} averaged over Bangladesh, from the ACAG satellite grids (van Donkelaar et al. 2021). Pollution exhibits a pronounced seasonal cycle, peaking in the dry winter months and falling sharply during the monsoon (June–September), motivating the inclusion of season-by-year fixed effects.

3.3 Thermal Inversion Data

With altitude, temperature decreases, with the air becoming colder at higher altitudes. However, a thermal inversion occurs when the reverse happens, that is, when temperature increases with an increase in altitude. In other words, a mass of cold air gets trapped below hot air due to its higher density, and traps pollutants beneath it (Arceo, Hanna, and Oliva 2016; Ye, Huang and Chen 2023; Zhou and Zhang 2023; Chan, Pelli and Vienne 2024). These contribute to increased pollution including particulate matter (Arceo et al. 2016; Jacobson and Jacobson 2002). Prior research also finds that severe thermal inversions are linked to some of the worst pollution events reported in history (Balakrishnan and Tsaneva 2021; Bailey et al. 2011; Malek et al. 2006).

I utilize the Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA2, version 5.12.4) to build thermal inversion measure. Produced by National Aeronautics and Space Administration (NASA) Global Modelling and Assimilation Office (GMAO), this dataset provides the latest version of global atmospheric reanalysis for 42 sea-

level pressure layers, ranging from 110 m to 36,000m. This data is collected every 3 hours and has a spatial resolution of $0.5^\circ \times 0.625^\circ$ (56x64 km). I define the occurrence of a thermal inversion when the temperature in the lower layer closest to the ground (975hPa-325m) is lower than that in the upper layer (925hPa-760m) (Ye, Huang, and Chen 2023; Chen, Wang, and Zhang 2022). I determine the average number of thermal inversions for each month at the household level. As a robustness check, I construct an alternative measure of inversions, inversion strength, which is determined by taking the temperature difference between the 975hPa and the 925hPa layers. This helps capture the magnitude of the inversions and can be either positive or negative in nature.

Figure 4 plots total monthly thermal inversions from 2011-20. Inversions exhibit strong seasonality, with higher number of inversions being experienced in dry winter months (October-March) ranging between 35-55 inversion counts per month. During the monsoon season, they fall to near zero counts. Together, the series highlights the temporal variation of thermal inversions across the years.

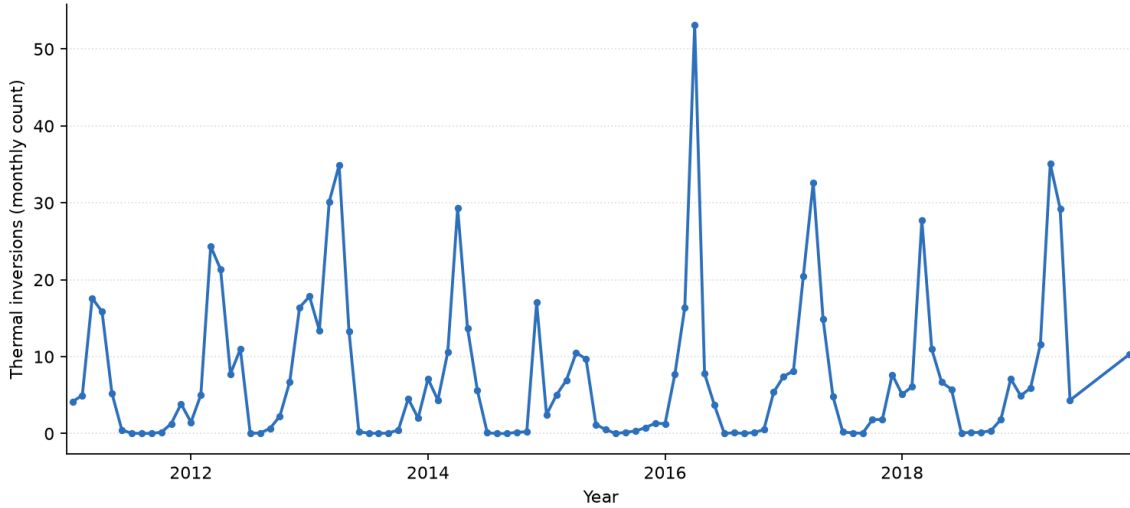


Figure 4: Monthly Thermal Inversions in Bangladesh, 2011–2019

Note: Monthly count of 925–975 hPa (inv4-2) thermal inversions averaged over Bangladesh, from MERRA-2 3-hourly temperature levels. Inversions exhibit strong seasonality, peaking in the dry winter and pre-monsoon months and falling to near zero during the monsoon — the same seasonal pattern as $PM_{2.5}$ (Figure 3), consistent with thermal inversions trapping pollution near the surface.

Figure 5 maps the monthly number of thermal inversions across the 3 BIHS survey waves.

Across most of the country, the frequency of inversions is fairly even and range between 0-40 inversions per month. Higher inversions are experienced in the southeastern parts of Bangladesh.

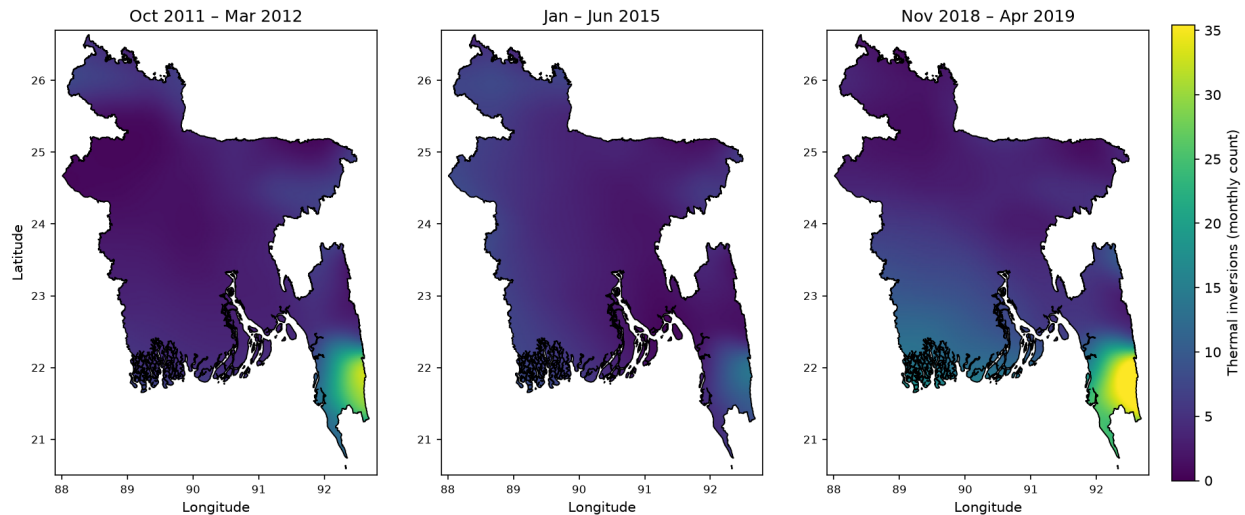


Figure 5: Average Thermal Inversions in Bangladesh (2011–2019)

Note: Average monthly count of 925–975 hPa (inv4-2) thermal inversions over each survey window — Oct 2011–Mar 2012, Jan–Jun 2015, and Nov 2018–Apr 2019 — from MERRA-2 3-hourly temperature levels, interpolated from the native reanalysis grid ($\sim 0.5^\circ$) to a continuous surface and clipped to the Bangladesh boundary. An inversion occurs when the 925 hPa temperature exceeds the 975 hPa temperature. Inversions are most frequent in the south-eastern hills; the scale is common across panels.

3.4 Weather Data

Information pertaining to weather controls affecting both air quality and workers' labor supply including wind speed, relative humidity, surface pressure, temperature and precipitation is taken from European Center for Medium-Range Weather Forecasts (ECMWF) Copernicus Climate Change Service (C3S) ERA5 reanalysis data. This provides information at a high spatial resolution of $0.25^\circ \times 0.25^\circ$ grid (28x28 km). I aggregate hourly data to determine monthly averages at the household level in the data.

Thermal inversions arise due to specific atmospheric conditions and may be closely associated with meteorological factors such as temperature and precipitation which also have an impact on human activities. To account for potential confounding arising from seasonal

weather patterns, I control for these variables (Arceo, Hanna, and Oliva 2016; Chen et al. 2017; Deschenes and Greenstone 2007; Ye, Huang and Chen 2023). This allows me to isolate the effect of thermal inversions on pollution from broader meteorological factors, alleviating any concerns regarding the exclusion restriction.

Table 1 represents the temporal distribution of the data used in this paper. I used all 3 rounds of Bangladesh Integrated Household Survey (BIHS). This first round includes data from households interviewed at the end of 2011 and the beginning of 2012 while the second round includes observations from the first two quarters of 2015. For the third round, responses are collected from the last quarter of 2018 and the first two quarters of 2019.

Table 1: Temporal Distribution of Data

Quarter	Year					Total
	2011	2012	2015	2018	2019	
Quarter 1 (Jan–Mar)	0	5,789	6,265	0	7,391	19,445
Quarter 2 (Apr–Jun)	0	0	5,672	0	1,267	6,939
Quarter 3 (Jul–Sep)	0	0	4	0	0	4
Quarter 4 (Oct–Dec)	5,502	0	0	1,320	0	6,822
Total	5,502	5,789	11,941	1,320	8,658	33,210

Table 2 presents descriptive statistics by wave for all variables used in the analysis. The pooled BIHS sample consists of 33210 individuals aged 15-64 years in 6147 households, of which 36% are observed in all three waves, 31% are observed in two waves and 33% are observed in only one wave. Across waves, hours worked range between 100-120 per month, with the average age of working individuals being 36-37 years. The average household size varies between 4.5 and 4.9 members, with nearly 80% of the sample being married across all 3 waves. Average pollution levels falls from 102.7 $\mu\text{g}/\text{m}^3$ to 63.4 $\mu\text{g}/\text{m}^3$ between 2011-12 and 2015 and then rises again in 2018-19 to 95.8 $\mu\text{g}/\text{m}^3$ on average. Similarly, the count of thermal inversions and ranges between 0-66 per month. Weather controls including rainfall, wind speed, air pressure relative humidity and temperature exhibit some variation but remain fairly stable across rounds.

Table 2: Descriptive Statistics by Survey Wave

Variable	2011-12					2015					2018-19				
	Obs	Mean	SD	Min	Max	Obs	Mean	SD	Min	Max	Obs	Mean	SD	Min	Max
Panel A: Working Hours															
Monthly Hours	11,291	116.070	92.930	2.167	364.000	11,941	109.241	94.689	0.433	364.000	9,978	103.222	94.198	0.433	364.000
Log Monthly Hours	11,291	4.322	1.041	0.773	5.897	11,941	4.216	1.081	-0.836	5.897	9,978	4.110	1.138	-0.836	5.897
Panel B: Demographics															
Age	11,291	36.289	12.232	15.000	64.000	11,941	37.361	12.761	15.000	64.000	9,978	37.146	13.043	15.000	64.000
Education	11,291	3.769	3.893	0.000	16.000	11,941	4.109	3.976	0.000	16.000	9,978	4.614	4.145	0.000	16.000
Household Size	11,291	4.510	1.626	1.000	15.000	11,941	4.821	1.842	1.000	20.000	9,978	4.892	1.952	1.000	19.000
Married	11,291	0.827	0.378	0.000	1.000	11,941	0.813	0.390	0.000	1.000	9,978	0.785	0.411	0.000	1.000
Male	11,291	0.527	0.499	0.000	1.000	11,941	0.501	0.500	0.000	1.000	9,978	0.485	0.500	0.000	1.000
Age group															
15-30	11,291	0.395	0.489	0.000	1.000	11,941	0.341	0.474	0.000	1.000	9,978	0.330	0.470	0.000	1.000
31-45	11,291	0.375	0.484	0.000	1.000	11,941	0.385	0.487	0.000	1.000	9,978	0.369	0.483	0.000	1.000
46-64	11,291	0.230	0.421	0.000	1.000	11,941	0.275	0.446	0.000	1.000	9,978	0.301	0.459	0.000	1.000
Income tercile															
Low	11,291	0.221	0.415	0.000	1.000	11,941	0.227	0.419	0.000	1.000	9,978	0.162	0.368	0.000	1.000
Middle	11,291	0.355	0.479	0.000	1.000	11,941	0.352	0.477	0.000	1.000	9,978	0.374	0.484	0.000	1.000
High	11,291	0.424	0.494	0.000	1.000	11,941	0.421	0.494	0.000	1.000	9,978	0.464	0.499	0.000	1.000
Panel C: Pollution, Thermal Inversions & Weather															
PM2.5	11,291	102.679	23.752	43.300	167.200	11,941	63.425	24.374	13.800	152.600	9,978	95.832	34.581	34.300	200.400
Thermal Inversions	11,291	1.762	4.952	0.000	47.000	11,941	4.260	4.550	0.000	44.000	9,978	5.113	6.515	0.000	66.000
Precipitation	11,291	12.090	17.705	0.023	111.664	11,941	175.954	184.962	1.273	942.744	9,978	54.824	54.328	0.000	311.337
Temperature	11,291	20.262	2.877	14.462	28.593	11,941	25.185	2.482	17.534	29.840	9,978	23.048	2.726	18.149	29.385
Wind Speed	11,291	1.808	0.363	0.828	5.191	11,941	2.211	0.473	0.965	5.606	9,978	2.042	0.359	1.083	4.452
Air Pressure	11,291	1010.352	6.585	934.983	1015.753	11,941	1008.128	7.227	931.727	1015.638	9,978	1009.924	7.764	934.246	1016.430
Relative Humidity	11,291	75.449	5.887	54.842	83.864	11,941	76.533	6.194	59.155	88.718	9,978	71.658	5.354	52.614	83.913

Note: Working sample of individuals aged 15–64 with a primary work activity (top-1%-trimmed monthly hours) and complete controls, pollution, instrument and weather; $N = 33,210$ (11,291 in 2011–12; 11,941 in 2015; 9,978 in 2018–19) across 6,147 households. Thermal inversions are the monthly count of 925–975 hPa (inv4-2) inversion occurrences. Male, age group, and income tercile are indicator variables; income terciles are of household labour income.

3.5 Statistical Challenges and Identification Strategy

Because pollution is not randomly assigned across space or time, it is difficult to determine the causal effect of pollution on hours worked. One concern is simultaneity where economic activities are correlated with both PM2.5 and hours worked (Yamada and Narita 2025). Areas with higher labor force participation and industrial activities experience higher pollution, thereby leading to a positive correlation between pollution and labor supply. Second, residential sorting can bias estimates, with individuals choosing where to live based on several factors including employment opportunities and local amenities available in that area (Freeman et al. 2019; Khanna et al. 2021; Zivin and Neidell 2013; Aguilar-Gomez et al. 2022; Mink 2025). Higher income households or more health-conscious workers will “vote with their feet” and sort into locations having better air quality (Chen, Lei, and Tan-Soo 2021; Chen, Oliva, and Zhang 2022; Freeman et al. 2019; Gu, Bian, and Elahi 2020; Ito and Zhang 2020). Conversely, cities serve as hubs for highly skilled workers due to better employment opportunities, yet these areas are also a major source of pollution (Zivin and Neidell 2013). However, such areas may also have better access to healthcare, which could circumvent some of the adverse effects of pollution. As a result, the direction of bias arising from endogeneity remains ambiguous.

Third, selection bias can introduce bias with workers adjusting their labor supply according to air quality deterioration (Ye, Huang and Chen 2023; Yu 2023). They may adopt avoidance behaviors including working less hours or working less efficiently or take up caregiving responsibilities when household members fall ill or avail medical insurance (Ye, Huang and Chen 2023). Measurement error also poses a serious challenge and can bias OLS estimates (Chan, Pelli, and Vienne 2023). Pollution monitors, especially in developing countries, are sparsely located and their distance from survey sites introduces measurement error in data which is difficult to account for (Ye, Huang, and Chen 2023). Compared to developed countries where the monitor density is much higher (US-3.4 monitors/million people), developing countries such as Bangladesh (0.07 monitors/million people) and India (0.14 mon-

itors/million people) have a scarcity of ground-level monitors (Brauer et al. 2019). Local data manipulation and misreporting pollution levels may also lead to inaccurate results (Gu, Bian and Elahi 2020).

To address these endogeneity concerns, I employ the instrumental variable (IV) method and exploit the meteorological phenomenon of thermal inversions. This occurs when a mass of cold air gets caught below a mass of hot air, trapping pollutants and degrading air quality (Arceo, Hanna, and Oliva 2016). Controlling for weather variables likely to influence working hours, the occurrence of thermal inversions appears to be as good as randomly assigned and independent of economic activities. Because of being random, they are unlikely to be related to socioeconomic conditions and therefore satisfy the exclusion restriction condition for an instrumental variable. Furthermore, I argue that thermal inversions affect labor supply solely through the channel of changing air quality levels.

Thermal inversions have been widely used as an instrumental variable for pollution in the environmental economics literature (Arceo, Hanna, and Oliva 2016; Balakrishnan and Tsaneva 2021; Chen et al. 2021; Jans et al. 2014; Sager 2019; Dechezlepretre, Rivers, and Stadler 2019; Fu et al. 2021; Ye, Huang, and Chen 2023; Molin 2021; Deschenes et al. 2020; Mink 2025). Other papers have used different instruments for pollution including inversion strength (Arceo, Hanna, and Oliva 2016; Chen, Oliva and Zhang 2017) changes in local wind direction (Han et al. 2023), maximum wind speed (Gu et al. 2019; Gu, Bian and Elahi 2020; Li and Zhang 2019; Kögel 2023), planetary boundary layer height (Mink 2025), temperature and precipitation (Kamble 2024), ventilation index (Broner, Bustos, and Carvalho 2012; He et al. 2019) and daily airplane taxi time (Schlenker and Walker 2016).

4 Empirical Analysis

I adopt an instrumental variable two stage least squares approach to determine the causal impact of pollution exposure on hours worked using thermal inversions as a source of exogenous

variation in pollution levels as below:

$$\text{PM2.5}_{g,t-1} = \alpha_0 + \alpha_1 \text{Thermal Inversions}_{g,t-1} + \alpha_2 X_{i,t} + \alpha_3 W_{g,t-1} + \rho_d + \gamma_{s,y} + \varepsilon_{i,t} \quad (1)$$

$$\log(\text{Hours}_{i,t}) = \beta_0 + \beta_1 \widehat{\text{PM2.5}}_{g,t-1} + \beta_2 X_{i,t} + \beta_3 W_{g,t-1} + \tau_d + \delta_{s,y} + \vartheta_{i,t} \quad (2)$$

where $\log(\text{Hours}_{i,t})$ denotes hours worked by individual i (in household h) in the reference month t . $\text{PM2.5}_{g,t-1}$ denotes the average monthly concentration of fine particulate matter (PM_{2.5}) in pollution grid cell g in month $t - 1$, that is, at a one-month lag relative to the labor reference period. $\text{Thermal Inversions}_{g,t-1}$ denotes the corresponding count of thermal inversions in cell g in the same lagged month $t - 1$, so that pollution and its instrument are both measured one month before labor outcomes are observed. $X_{i,t}$ represents individual and household controls including age, education, household size and marital status. $W_{g,t-1}$ represent one-month-lag monthly weather controls including temperature, precipitation, wind speed, air pressure, and relative humidity. ρ_d and τ_d represent district fixed effects that absorb time-invariant socioeconomic and geographic factors specific to each district. $\gamma_{s,y}$ and $\delta_{s,y}$ represent season-by-year fixed effects—where s indexes the season and y the survey year (2011, 2012, 2015, 2018, and 2019)—that absorb season-specific shocks to pollution and labor supply within each wave, common to all areas. I use these fixed effects instead of household fixed effects as the panel is highly unbalanced and using them would discard much of the sample. $\varepsilon_{i,t}$ and $\vartheta_{i,t}$ are the error terms. Standard errors are clustered at the pollution-grid level g .

Equation 1 represents the first stage estimation where the coefficient of interest is α_1 . This represents the change in pollution levels attributable to a change in thermal inversions. Equation 2 represents the second stage regression where β_1 captures the impact on labor supply of change in monthly hours worked.

Table 3 reports results from IV FE regression. Column 1 reports the first stage relationship between pollution and thermal inversions. This is positive and statistically significant, with a one standard deviation increase in thermal inversions being associated with a 2 $\mu\text{g}/\text{m}^3$

increase in PM2.5 concentrations, corresponding to approximately 2.2% of mean pollution levels ($94.1 \mu\text{g}/\text{m}^3$). The Kleibergen-Paap F statistic from the first stage regression is 39.79, greater than the Stock-Yogo 10% maximal IV size critical value of 16.38 (Stock and Yogo 2005).

Column (2) reports the second-stage IV estimates. The estimated effect of pollution on hours worked is negative and statistically significant at the 1% level. The point estimate implies that a $10 \mu\text{g}/\text{m}^3$ increase in PM2.5 is associated with a 18.8% reduction in monthly hours worked, a loss of roughly 20.7 hours per month relative to the sample mean of 109.8 hours. Figure A2 shows the first stage and reduced form graphically.

The magnitude of the point estimate is larger than results found in prior literature where a $10 \mu\text{g}/\text{m}^3$ increase in PM2.5 corresponds to an approximately 3-6% reduction in weekly hours worked in developing country settings. Similar studies have found a decrease in labor supply in developing countries (Aragon et al. 2017; Kim, Manley and Radoias 2017; Fan and Grainger 2023). The larger effect is plausibly related to low-income, rural households exposed to very high ambient pollution levels. Furthermore, poorer baseline health and lack of adequate healthcare exacerbate pollution-induced health impacts (Balakrishnan and Tsaneva 2021).

Table 3: Impact of PM_{2.5} on Labor Supply

	(1) PM _{2.5} <i>First Stage</i>	(2) ln(Hours) <i>Second Stage</i>
Thermal Inversions	101.3460*** (16.0674)	
PM _{2.5}		−0.0188*** (0.0063)
District FE	Yes	Yes
Season × Year FE	Yes	Yes
Observations	33,210	33,210
R^2	0.849	—
N clusters	4,134	4,134
First-stage F	39.79	

Notes. Instrument: thermal-inversion count $inv4_2 = T(925) - T(975)$, one-month lag. Column (1) reports the first-stage regression of PM_{2.5} on the thermal-inversion count (Equation 1); Column (2) reports the second-stage IV estimate of log monthly hours on instrumented PM_{2.5} (Equation 2). Individual (age, education, household size, marital status) and weather controls (precipitation, temperature, wind speed, surface pressure, relative humidity; one-month lag) are included in both stages. All specifications include district and season × year fixed effects. Standard errors, in parentheses, are clustered at the pollution grid level (0.01-degree ACAG cell). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.1 Heterogeneity

In this section, I focus on the heterogeneous impact of pollution on labor supply across various demographic, socioeconomic and occupational groups. For this, I allow the effect of pollution exposure to vary with individual and household characteristics. PM_{2.5} and its interaction with the moderator are both treated as endogenous and instrumented by thermal inversions and their interaction with the moderator, giving the two first-stage equations:

$$\begin{aligned} \text{PM2.5}_{g,t-1} = & \pi_0 + \pi_1 \text{Thermal Inversions}_{g,t-1} + \pi_2 (\text{Thermal Inversions}_{g,t-1} \times M_i) \\ & + \pi_3 M_i + \pi_4 X_{i,t} + \pi_5 W_{g,t-1} + \rho_d + \gamma_{s,y} + \varepsilon_{i,t} \end{aligned} \quad (3)$$

$$\begin{aligned} \text{PM2.5}_{g,t-1} \times M_i = & \omega_0 + \omega_1 \text{Thermal Inversions}_{g,t-1} + \omega_2 (\text{Thermal Inversions}_{g,t-1} \times M_i) \\ & + \omega_3 M_i + \omega_4 X_{i,t} + \omega_5 W_{g,t-1} + \rho_d + \gamma_{s,y} + \eta_{i,t} \end{aligned} \quad (4)$$

The second stage is:

$$\log(\text{Hours}_{i,t}) = \beta_0 + \beta_1 \widehat{\text{PM2.5}}_{g,t-1} + \beta_2 (\widehat{\text{PM2.5}}_{g,t-1} \times M_i) + \beta_3 M_i + \beta_4 X_{i,t} + \beta_5 W_{g,t-1} + \tau_d + \delta_{s,y} + \vartheta_{i,t} \quad (5)$$

where M_i denotes the moderator (e.g., an indicator for male, an age band, or an income tercile), and all other terms— $\text{Thermal Inversions}_{g,t-1}$, $X_{i,t}$, $W_{g,t-1}$, the district fixed effects ρ_d and τ_d , and the season-by-year fixed effects $\gamma_{s,y}$ and $\delta_{s,y}$ —are as defined in the baseline specification. The coefficient of interest is β_2 , which captures the differential effect of pollution on labor supply for individuals with $M_i = 1$ relative to the omitted group, while β_1 gives the effect for the base category. For categorical moderators (age band and income tercile), M_i is replaced by a vector of indicators, with one endogenous interaction and one instrument interaction per non-base category. Standard errors are clustered at the pollution-grid level g .

Table 4 reports heterogeneity estimates in the effect of pollution on labor supply, with each column interacting instrumented pollution with a single moderator. Column 2 examines heterogeneity by gender. For women, the omitted category, a $10 \mu\text{g}/\text{m}^3$ increase in PM2.5 reduces monthly hours worked by approximately 16% , equivalent to about 18 fewer hours worked per month. The interaction with the male indicator is positive and statistically insignificant. Column 3 reports heterogeneity by age. Relative to the youngest workers (aged 15–30), whose response is negative and statistically significant (-0.0176 , $p < 0.01$), the effect of pollution is significantly more negative for older workers, with the interaction for workers aged 46–64 being -0.0028 ($p < 0.10$). The estimate for workers aged 31–45 is not

statistically significant. These estimates indicate heightened sensitivity among older workers to pollution-induced health burden (Currie 2005; Arceo, Hanna and Oliva 2016).

Column 4 reports the heterogeneity by household income. For low income households, the effect is negative and statistically significant (-0.0153 , $p < 0.05$). However, the interactions for higher income households are not statistically significant. This provides evidence of there being no statistically significant income gradient in the labor supply response. Figure A3 plots out the heterogeneity estimates.

Table 4: Impact of PM_{2.5} on Labor Supply: Heterogeneity Estimates

	(1) Baseline	(2) Gender	(3) Age	(4) Income
PM _{2.5}	-0.0188*** (0.0063)	-0.0160*** (0.0054)	-0.0176*** (0.0064)	-0.0153** (0.0061)
PM _{2.5} × Male		0.0013 (0.0010)		
PM _{2.5} × Age 31–45			-0.0017 (0.0015)	
PM _{2.5} × Age 46–64			-0.0028* (0.0016)	
PM _{2.5} × Middle income				0.0015 (0.0016)
PM _{2.5} × High income				0.0008 (0.0017)
Controls	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Season × Year FE	Yes	Yes	Yes	Yes
Observations	33,210	33,210	33,210	33,210

Notes: The dependent variable is log monthly working hours. Instrumented PM_{2.5} (inv4_2, one-month lag) is interacted with each moderator; the interaction is instrumented by inversion × moderator. The omitted base categories are female (column 2), age 15–30 (column 3), and the low-income tercile (column 4); income terciles are of household labour income. Controls comprise age, education, household size, marital status, and one-month-lag precipitation, temperature, wind speed, surface pressure and relative humidity. All specifications include district and season × year fixed effects. Standard errors clustered at the pollution grid level (0.01-degree ACAG cell) in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.2 Potential Mechanisms

Prior literature documents several mechanisms through which pollution affects labor supply. Aragon et al. (2017) finds evidence of increased demand for caregiving in susceptible households a plausible mechanism with a $10 \mu\text{g}/\text{m}^3$ increase in PM_{2.5} leading to 2 less hours worked per week. Kim, Manley and Radoias (2017) find caregiving to be an important channel in the medium term, with women working 2.46 hours less in households having dependents as compared to men who reduce their labor supply by 1.17 hours. In the long term, however, own health serves as the underlying channel. Huang et al. (2021) find health to be a potential mechanism affecting labor supply-a $1 \mu\text{g}/\text{m}^3$ increase in pollution leads to the probability of being attacked by cardio cerebrovascular or respiratory diseases by 6.34%. Hoffman and Rud (2024) find that workers reduce their labor supply on high pollution days as avoidance behavior. Fan and Grainger (2023) find that environmentally aware workers are more likely to adopt avoidance behavior and reduce their labor supply by 0.26 hours (16 minutes) in response to a one unit increase in pollution levels. While my research focuses on determining reduced form labor supply responses, disentangling these channels remains an avenue for future research.

4.3 Robustness Checks

I conduct a series of robustness checks. These are presented in table 5. Column 1 reports baseline results while the subsequent columns examine the sensitivity of results by using alternative specifications. Column 1 reports estimate from the baseline specification. Columns 2 and 3 vary the meteorological controls, with column 2 dropping them altogether (-0.0080) and the latter including polynomial weather controls instead (-0.0309). The estimates remain negative and statistically significant in both cases. Column 4 uses a modified definition of thermal inversions which are now instead calculated as the total number of monthly thermal inversions. The coefficient remains negative and significant (-0.0218). In Column 5, I reconstruct thermal inversions by using the difference in air temperature between 950hPa and

975hPa pressure levels. My estimate remains negative and statistically significant (-0.0172). In Columns 6 and 7, I restrict the sample, first to having a balanced sample having the same households observed in all 3 waves (-0.0133) and then to prime working age individuals (aged 25-55 years) only (-0.0173). In Column 8, I conduct a placebo test by using future pollution and thermal inversions. Since these are observed after the period in which working hours are reported, they cannot impact labor supply. I find that the estimate is small, positive and not statistically significant, consistent with this explanation (0.0117). In the last two columns, I winsorize first the pollution variable and then working hours. In both cases, my coefficient remains similar to that estimated in the baseline regression. Overall, these robustness checks confirm that the findings are stable, and the main estimates are robust across alternative specifications. Figure A4 plots the robustness estimates across specifications.

Table 5: Impact of PM2.5 on Labor Supply: Robustness Checks

	(1)	(2)	(3)	(4)	(5)
	Baseline	No weather	Flexible weather controls	IV: total inversion count	Alternative Definition Thermal Inversions
PM2.5	−0.0188*** (0.0063)	−0.0080*** (0.0028)	−0.0309** (0.0139)	−0.0218*** (0.0073)	−0.0172*** (0.0065)
First-stage F	39.7	88.4	11.6	32.8	31.5
Weather controls	Continuous	None	Quadratic	Continuous	Continuous
Fixed effects	District	District	District	District	District
	+ Season × Year	+ Season × Year	+ Season × Year	+ Season × Year	+ Season × Year
SE clustering	Grid	Grid	Grid	Grid	Grid
Observations	33,210	33,210	33,210	33,210	33,210
Clusters	4,134	4,134	4,134	4,134	4,134
	(6)	(7)	(8)	(9)	(10)
	Balanced panel	Prime age (25–55)	Placebo: future pollution	Winsorized PM2.5	Winsorized hours
PM2.5	−0.0133** (0.0052)	−0.0173*** (0.0065)	0.0117 (0.0089)	−0.0197*** (0.0066)	−0.0186*** (0.0062)
First-stage F	48.6	37.7	20.7	37.8	39.7
Weather controls	Continuous	Continuous	Future	Continuous	Continuous
Fixed effects	District	District	District	District	District
	+ Season × Year	+ Season × Year	+ Season × Year	+ Season × Year	+ Season × Year
SE clustering	Grid	Grid	Grid	Grid	Grid
Observations	26,581	23,924	33,210	33,210	33,210
Clusters	2,980	4,012	4,134	4,134	4,134

Notes. Each column is a separate 2SLS regression of log monthly hours worked on PM2.5 (one-month lag), instrumented by thermal inversions. Column (1) is the baseline: average $\text{inv4_2} = T(925) - T(975)$ inversion count instrument, individual controls (age, education, household size, married), continuous one-month-lag weather (precipitation, temperature, wind, surface pressure, relative humidity), district and season-by-year fixed effects, and standard errors clustered at the pollution grid (0.01-degree van Donkelaar cell). Column (2) drops weather; (3) uses second-order polynomial (quadratic) weather; (4) uses total (rather than average) inversion count as the instrument; (5) replaces the instrument with an alternative inversion definition, $\text{inv3_2} = T(950) - T(975)$; (6) restricts to households observed in all three waves (balanced panel); (7) restricts to prime-age individuals 25–55; (8) is a placebo using interview-month pollution and inversions, which are future relative to the labor reference month and therefore cannot have caused the hours reported for it; (9)–(10) winsorize PM2.5 and hours at the 1st/99th percentiles. First-stage F is the cluster-robust Kleibergen–Paap F on the excluded instrument. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5 Conclusion

This paper examines the impact of fine particulate matter on labor supply in rural Bangladesh. Using thermal inversions as an instrument for pollution exposure, I estimate a negative effect of PM2.5 on monthly hours worked, with a $10 \mu\text{g}/\text{m}^3$ increase in PM2.5 associated with a reduction in monthly working hours of approximately 18.8%. The effect is stronger for older

workers although it does not differ significantly by gender or household income.

It is also important to note that the use of satellite-based pollution estimate, as is done in this paper, is prone to result in downward bias at higher pollution levels due to which these results are likely underestimating the true impact of pollution on labor supply (Fowlie, Knittel and Wolfram 2012). Several limitations of the analysis point to avenues for future research. The adoption of stronger or alternative instruments can help sharpen inference. Furthermore, improving the flexibility of the IV framework to accommodate non-linear and heterogeneous effects of pollution on working hours remains a further direction. Findings from this paper suggest that air pollution poses high labor market costs in low-income, developing countries such as Bangladesh.

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Appendix A

Table A1: Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Panel A: Working Hours					
Monthly Hours	33,210	109.754	94.084	0.433	364.000
Log Monthly Hours	33,210	4.220	1.088	-0.836	5.897
Panel B: Demographics					
Age	33,210	36.932	12.679	15	64
Education	33,210	4.145	4.014	0	16
Household Size	33,210	4.736	1.814	1	20
Married	33,210	0.809	0.393	0	1
Male	33,210	0.505	0.500	0	1
<i>Age group</i>					
15–30	33,210	0.356	0.479	0	1
31–45	33,210	0.377	0.485	0	1
46–64	33,210	0.268	0.443	0	1
<i>Income tercile</i>					
Low	33,210	0.205	0.417	0	1
Middle	33,210	0.359	0.478	0	1
High	33,210	0.435	0.494	0	1
Panel C: Pollution, Thermal Inversions & Weather					
PM2.5	33,210	86.508	32.731	13.800	200.400
Thermal Inversions	33,210	3.667	5.525	0	66
Precipitation	33,210	83.848	135.454	0.000	942.744
Temperature	33,210	22.869	3.393	14.462	29.840
Wind Speed	33,210	2.023	0.439	0.828	5.606
Surface Pressure	33,210	1009.424	7.253	931.727	1016.430
Relative Humidity	33,210	74.700	6.194	52.614	88.718

Note: The sample comprises 33,210 individual–month observations from the working sample (individuals aged 15–64 with a primary work activity and complete controls). Thermal inversions are the monthly count of 925–975 hPa (inv4-2) inversion occurrences. Male, age group, and income tercile are indicator variables; income terciles are of household labour income computed within survey wave.

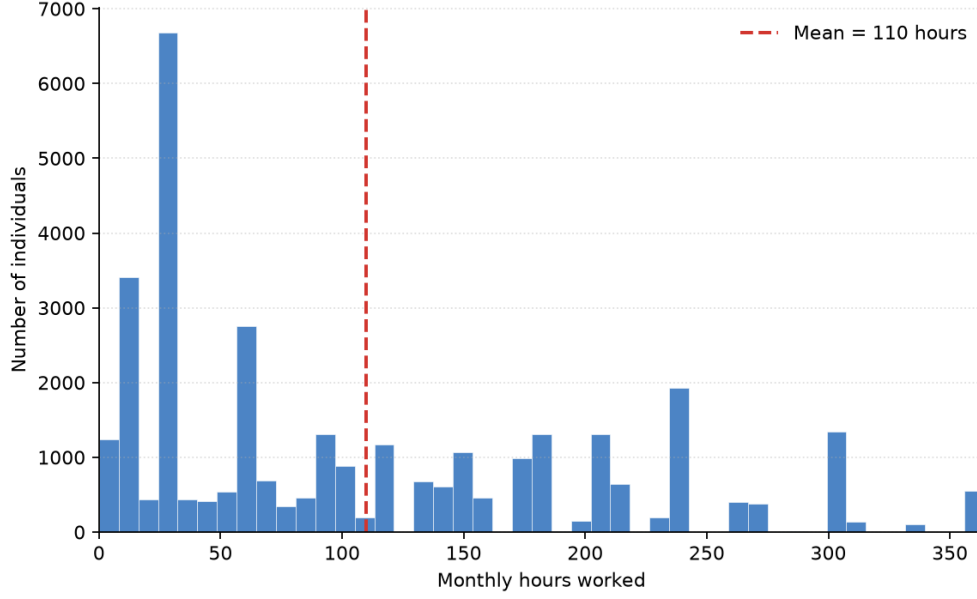


Figure A1: Distribution of Monthly Hours Worked

Note: Distribution of monthly hours worked (primary activity, weekly hours scaled by 52/12) for the 33,210 working-sample individuals, trimmed at the top 1%. The dashed line marks the mean (110 hours). The distribution is heavily right-skewed, with most individuals working relatively few hours per month.

Table A2: First Stage, Reduced Form, and 2SLS Estimates

	(1) First stage PM _{2.5}	(2) Reduced form ln(Hours)	(3) 2SLS ln(Hours)
Thermal inversions	101.346*** (16.067)	−1.907*** (0.541)	
PM _{2.5} (instrumented)			−0.0188*** (0.0063)
Controls	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Season × Year FE	Yes	Yes	Yes
Observations	33,210	33,210	33,210
First-stage F	39.79		

Notes. Column (1) regresses PM_{2.5} (one-month lag) on the thermal-inversion instrument; column (2) regresses log monthly hours directly on the instrument (reduced form); column (3) is the 2SLS estimate of log hours on instrumented PM_{2.5}. All columns include individual and one-month-lag weather controls, district and season-by-year fixed effects, and standard errors clustered at the pollution grid (0.01-degree ACAG cell). The 2SLS estimate equals the ratio of the reduced-form to the first-stage coefficient (−1.907/101.346 = −0.0188). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

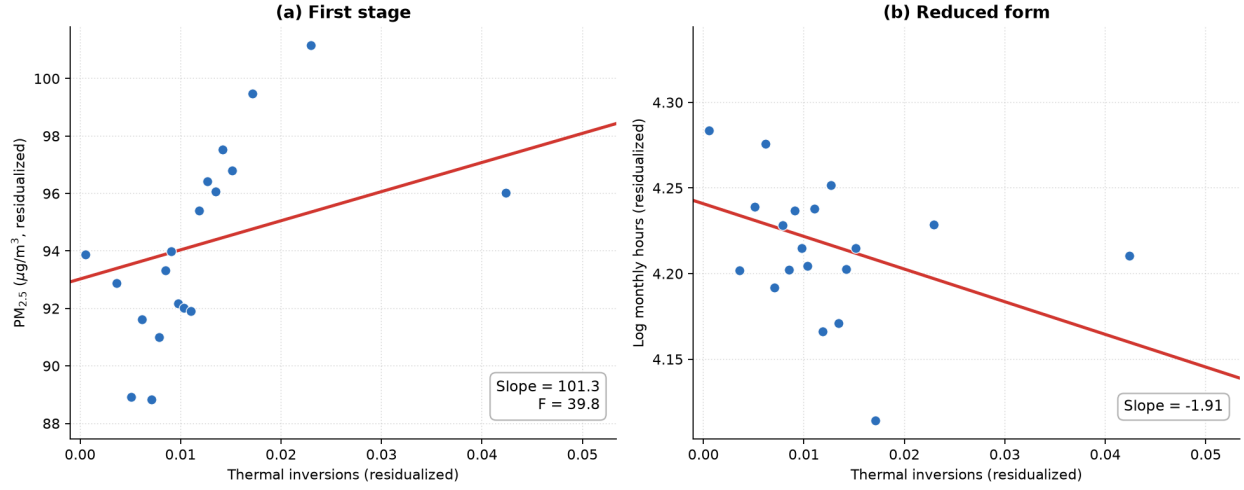


Figure A2: First Stage and Reduced Form

Note: Panel (a) plots $PM_{2.5}$ against the thermal-inversion instrument; panel (b) plots log monthly hours worked against the same instrument. In both panels the variables are residualized on the baseline controls (age, education, household size, marital status, one-month-lag weather) and district and season-by-year fixed effects; points are 20 equal-sized bins and the red lines are OLS fits, estimated on the 33,210 working-sample observations. Thermal inversions raise pollution (panel a; slope 101.3, Kleibergen-Paap $F = 39.8$) and reduce hours worked (panel b; slope -1.91). The ratio of the two slopes, $-1.91/101.3 = -0.0188$, equals the 2SLS estimate in Table 3.

Table A3: Impact of PM2.5 on Labor Supply

VARIABLES	(1) PM _{2.5}	(2) ln(Hours)
Thermal Inversions	101.346*** (16.067)	
PM _{2.5}		−0.0188*** (0.0063)
Age	0.0112 (0.00702)	0.00708*** (0.000634)
Education	0.0556** (0.0262)	−0.000975 (0.00209)
Household Size	0.0418 (0.107)	0.0185*** (0.00495)
Married	−0.0351 (0.228)	−0.0731*** (0.0203)
Precipitation	0.000381 (0.00269)	0.000499*** (0.000118)
Temperature	−4.342*** (0.136)	−0.0968*** (0.0282)
Wind Speed	5.498*** (0.882)	0.124** (0.0541)
Air Pressure	0.577*** (0.0346)	0.0121*** (0.00440)
Relative Humidity	0.428*** (0.0824)	0.00361 (0.00452)
Constant	−444.1 (34.22)	−4.803 (3.734)
District FE	Yes	Yes
Season × Year FE	Yes	Yes
Observations	33,210	33,210
R^2	0.849	—
N clusters	4,134	4,134
First-stage F	39.79	

Standard errors clustered at the pollution grid (0.01-degree ACAG cell) in parentheses. Instrument: thermal-inversion count (inv4-2, one-month lag). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A4: Impact of PM_{2.5} on Labor Supply: Heterogeneity Estimates (Full Controls)

	(1) Baseline	(2) Gender	(3) Age	(4) Income
PM _{2.5}	-0.0188*** (0.0063)	-0.0160*** (0.0054)	-0.0176*** (0.0064)	-0.0153** (0.0061)
× Male		0.0013 (0.0010)		
× Age 31–45			-0.0017 (0.0015)	
× Age 46–64			-0.0028* (0.0016)	
× Middle income				0.0015 (0.0016)
× High income				0.0008 (0.0017)
Male		1.3001*** (0.0985)		
Age 31–45			0.4070*** (0.1369)	
Age 46–64			0.4576*** (0.1504)	
Middle income				0.3694** (0.1518)
High income				0.6154*** (0.1569)
Age	0.0071*** (0.0006)	0.0016*** (0.0005)		0.0078*** (0.0006)
Education	-0.0010 (0.0021)	-0.0143*** (0.0017)	-0.0015 (0.0021)	-0.0040** (0.0020)
Household Size	0.0185*** (0.0050)	-0.0163*** (0.0044)	0.0173*** (0.0049)	-0.0207*** (0.0050)
Married	-0.0731*** (0.0203)	0.1304*** (0.0179)	-0.0986*** (0.0203)	-0.0964*** (0.0192)
Precipitation	0.0005*** (0.0001)	0.0004*** (0.0001)	0.0005*** (0.0001)	0.0005*** (0.0001)
Temperature	-0.0968*** (0.0282)	-0.0841*** (0.0238)	-0.0978*** (0.0280)	-0.0748*** (0.0255)
Wind Speed	0.1238** (0.0542)	0.0723 (0.0445)	0.1254** (0.0539)	0.0890* (0.0480)
Air Pressure	0.0121*** (0.0044)	0.0104*** (0.0037)	0.0122*** (0.0044)	0.0112*** (0.0039)
Relative Humidity	0.0036 (0.0045)	-0.0013 (0.0038)	0.0038 (0.0045)	0.0031 (0.0041)
District FE	Yes	Yes	Yes	Yes
Season × Year FE	Yes	Yes	Yes	Yes
Observations	33,210	33,210	33,210	33,210
Clusters	4,134	4,134	4,134	4,134

Notes: Full-controls version of the heterogeneity estimates (Table 4). The dependent variable is log monthly working hours. Instrumented PM_{2.5} (inv4-2, one-month lag) and its interaction with each moderator are treated as endogenous and instrumented by the thermal-inversion count and its interaction with the moderator. Omitted base categories: female (column 2), age 15–30 (column 3), and the low-income tercile (column 4); continuous age is dropped in column 3 as it is collinear with the age bands. All columns include district and season × year fixed effects. Standard errors clustered at the pollution grid (0.01-degree ACAG cell) in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

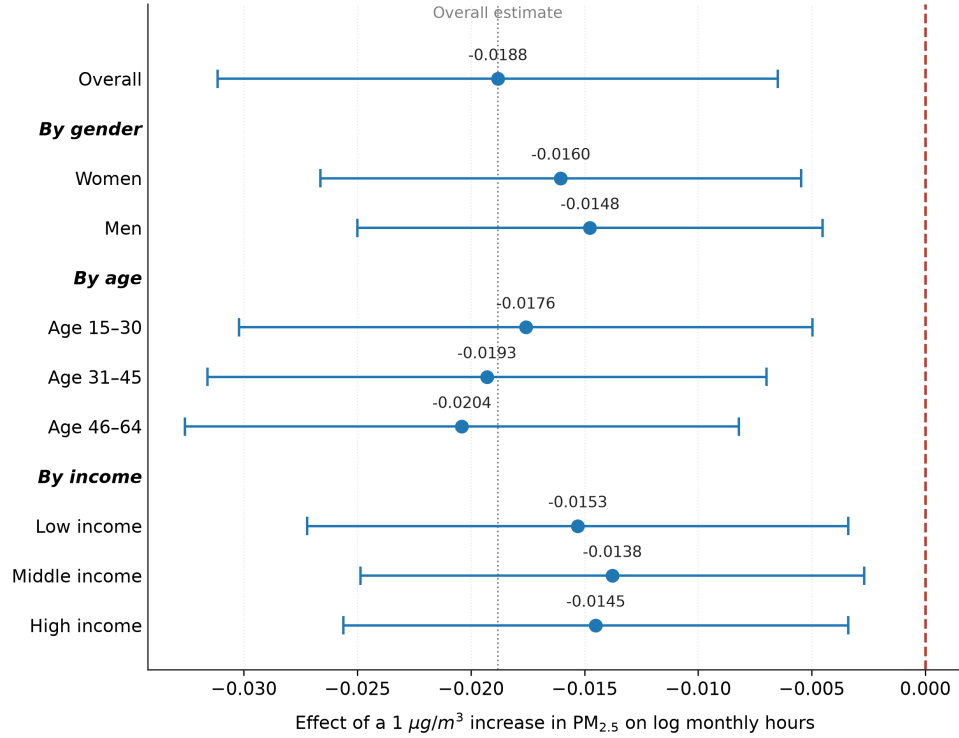


Figure A3: Heterogeneity in the Effect of PM2.5 on Labor Supply

Note: Each marker is the implied 2SLS effect of a $1 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ on log monthly hours for the indicated group, with 95% confidence intervals (horizontal bars). Subgroup effects are the sum of the base coefficient and the relevant interaction from the corresponding column of Table 4; standard errors for the combined effects account for the covariance between the two terms. The red dashed line is zero; the grey dotted line is the overall estimate (-0.0188). Standard errors are clustered at the pollution grid.

Table A5: Robustness of the Effect of PM_{2.5} on Log Monthly Hours (Full Controls)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Baseline	No weather	Poly. weather	Total inv.	Alternative Definition Thermal Inversions	Balanced	Prime age	Placebo	Winsor PM	Winsor hrs
PM _{2.5}	-0.0188*** (0.0063)	-0.0080*** (0.0028)	-0.0309** (0.0139)	-0.0218*** (0.0073)	-0.0172*** (0.0065)	-0.0133** (0.0052)	-0.0173*** (0.0065)	0.0117 (0.0089)	-0.0197*** (0.0066)	-0.0180*** (0.0062)
Age	0.0071*** (0.0006)	0.0070*** (0.0006)	0.0071*** (0.0007)	0.0071*** (0.0006)	0.0071*** (0.0006)	0.0079*** (0.0007)	0.0053*** (0.0009)	0.0068*** (0.0006)	0.0071*** (0.0006)	0.0070*** (0.0006)
Education	-0.0010 (0.0021)	-0.0015 (0.0020)	-0.0015 (0.0022)	-0.0008 (0.0021)	-0.0011 (0.0021)	-0.0007 (0.0024)	0.0141*** (0.0023)	-0.0023 (0.0020)	-0.0009 (0.0021)	-0.0008 (0.0021)
Household Size	0.0185*** (0.0050)	0.0174*** (0.0043)	0.0200*** (0.0054)	0.0187*** (0.0052)	0.0184*** (0.0049)	0.0123** (0.0053)	0.0174*** (0.0053)	0.0178*** (0.0043)	0.0180*** (0.0050)	0.0188*** (0.0049)
Married	-0.0731*** (0.0203)	-0.0722*** (0.0199)	-0.0754*** (0.0210)	-0.0732*** (0.0205)	-0.0730*** (0.0202)	-0.0702*** (0.0229)	-0.0094 (0.0308)	-0.0721*** (0.0196)	-0.0732*** (0.0203)	-0.0760*** (0.0200)
Precipitation	0.0005*** (0.0001)	0.0026** (0.0011)	0.0026** (0.0011)	0.0005*** (0.0001)	0.0005*** (0.0001)	0.0013*** (0.0003)	0.0005*** (0.0001)	0.0005*** (0.0001)	0.0005*** (0.0001)	0.0005*** (0.0001)
Temperature	-0.0968*** (0.0282)	1.1067** (0.4687)	1.1067** (0.4687)	-0.1097*** (0.0327)	-0.0897*** (0.0294)	-0.0774*** (0.0246)	-0.0880*** (0.0291)	0.0614 (0.0402)	-0.1002*** (0.0296)	-0.0957*** (0.0279)
Wind Speed	0.1238** (0.0542)	0.6282* (0.3675)	0.6282* (0.3675)	0.1423** (0.0597)	0.1138** (0.0537)	0.0662 (0.0612)	0.1288** (0.0541)	-0.1335*** (0.0349)	0.1214** (0.0540)	0.1218** (0.0537)
Air Pressure	0.0121*** (0.0044)	-0.9739 (0.5925)	-0.9739 (0.5925)	0.0139*** (0.0050)	0.0111** (0.0045)	0.0084** (0.0038)	0.0123*** (0.0046)	-0.0040 (0.0036)	0.0126*** (0.0046)	0.0120*** (0.0044)
Relative Humidity	0.0036 (0.0045)	0.4760** (0.2299)	0.4760** (0.2299)	0.0049 (0.0049)	0.0029 (0.0044)	-0.0009 (0.0044)	0.0037 (0.0049)	-0.0074*** (0.0025)	0.0038 (0.0046)	0.0034 (0.0045)
Weather controls	Cont.	None	Quad.	Cont.	Cont.	Cont.	Cont.	Future	Cont.	Cont.
Fixed effects	D+S×Y	D+S×Y	D+S×Y	D+S×Y	D+S×Y	D+S×Y	D+S×Y	D+S×Y	D+S×Y	D+S×Y
First-stage F	39.7	88.4	11.6	32.8	31.5	48.6	37.7	20.7	37.8	39.7
Observations	33,210	33,210	33,210	33,210	33,210	26,581	23,924	33,210	33,210	33,210
Clusters	4,134	4,134	4,134	4,134	4,134	2,980	4,012	4,134	4,134	4,134

Notes. Full-controls version of Table 5. Each column is a separate 2SLS regression of log monthly hours on PM_{2.5} (one-month lag), instrumented by thermal inversions. All columns include individual controls (age, education, household size, married), district and season $\times$ year fixed effects ("D+S×Y"), and standard errors clustered at the pollution grid (0.01-degree ACAG cell). Weather controls are one-month-lag and continuous ("Cont.") unless noted: column 2 drops them, column 3 adds quadratic terms ("Quad."); the squared terms are included but not shown), and column 8 (placebo) uses interview-month weather ("Future"). Column 4 instruments with the total inversion count. Column 5 replaces the instrument with the alternative inversion definition $\text{inv3}_2 = T(950) - T(975)$. Columns 6–7 restrict the sample, column 8 uses interview-month pollution and inversions (future relative to the labor reference month), and columns 9–10 winsorize PM_{2.5}/hours at 1/99. First-stage F is the cluster-robust Kleibergen-Paap F. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

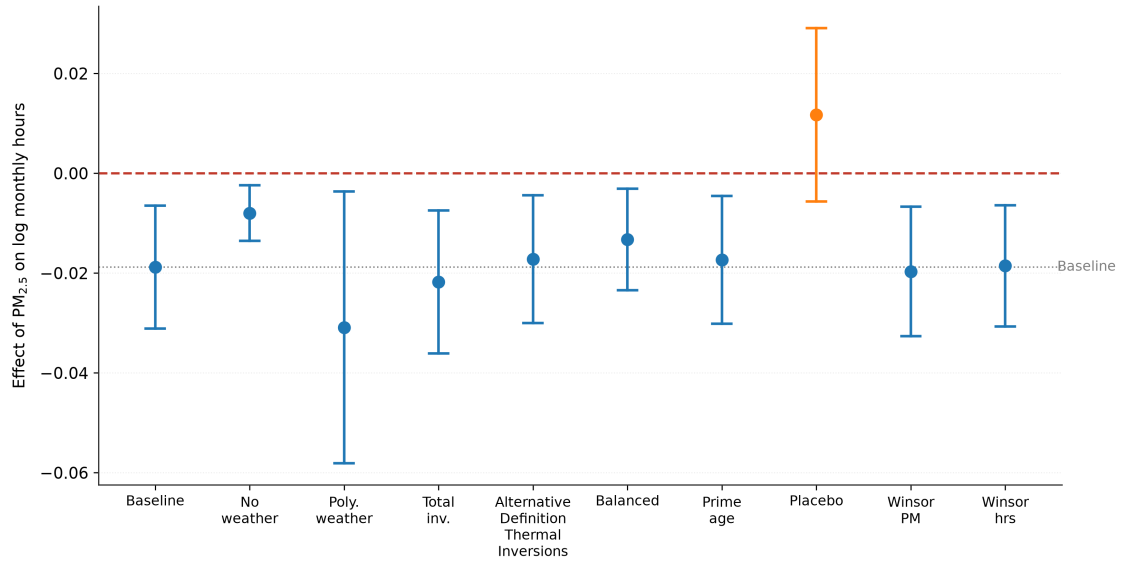


Figure A4: Robustness of the Effect of PM2.5 on Labor Supply

Note: Each marker is the 2SLS estimate of the effect of PM_{2.5} on log monthly hours from the corresponding column of Table 5, with 95% confidence intervals. The red dashed line is zero; the grey dotted line is the baseline estimate (−0.0188). The placebo (orange), which uses future rather than lagged pollution and inversions, is centered near zero with a confidence interval that includes zero. All other specifications yield negative, statistically significant estimates clustered around the baseline. Standard errors are clustered at the pollution grid.